

Quantifying Threshold Manipulation in the Presence of Rounding: The Case of Lead Monitoring in US Drinking Water[†]

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Many laws and economic actions depend on thresholds. As a consequence, threshold manipulation is a common concern in a variety of settings. Existing methods for detecting and quantifying threshold manipulation assume a continuous counterfactual distribution absent manipulation. This assumption is violated in the presence of rounding, which is prevalent in many applications and distinct from manipulation. This paper develops methods for testing and quantifying threshold manipulation when rounding is a prominent feature of the data. We demonstrate the usefulness of our approach in an empirical application examining threshold manipulation in lead monitoring under the US Safe Drinking Water Act. (JEL C24, I28, K32, Q53, Q58)

Economists have documented a range of settings where agents respond to policy thresholds.¹ The literature has correspondingly developed several methods to test for threshold manipulation (e.g., McCrary 2008), quantify threshold manipulation (e.g., Ghanem, Shen, and Zhang 2020; Diamond and Persson 2016), and quantify bunching (e.g., Chetty et al. 2011; Bertanha, McCallum, and Seeger 2023). Importantly, these methods assume that the counterfactual distribution is continuous. Rounding, which is common in many applications, introduces discontinuities, even in the absence of manipulation.

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¹We use the term *policy threshold* to refer to thresholds at which treatment, status, or other incentives change.

This paper develops methods to detect and quantify threshold manipulation in settings where rounding is a prominent data feature.² We first augment the potential outcomes framework to account for rounding and threshold manipulation. Using this framework, we demonstrate that existing manipulation measures are upwardly biased in the presence of rounding. We then develop a new manipulation test that relies on the insight that rounding should only move mass within a “roundable” interval around policy thresholds. Therefore, absent manipulation, the proportion of values in this interval should be equal in the observed and counterfactual distributions. We also develop a new manipulation measure that we call “excess rounding.” This measure captures the difference between the proportion of values equal to the threshold in the observed distribution and the counterfactual distribution absent manipulation. Identifying this counterfactual requires quantifying rounding absent manipulation. We do so by leveraging rounding behavior in parts of the distribution that are not thought to be influenced by manipulation behavior.

We demonstrate the advantages of our methods relative to existing approaches with an empirical application and a simulation study calibrated to the application. Our empirical application examines the accuracy of lead reporting in US drinking water from 2011 to 2020. As with nearly all water and air quality standards, the US Environmental Protection Agency (EPA) relies on public water systems (PWSs) to self-report performance metrics as part of their compliance process. These reports determine whether PWSs face costly monitoring and remediation requirements under the Lead and Copper Rule (LCR). Meeting certain thresholds exempts PWSs from these costly requirements, providing an incentive to game the regulation by manipulating data around key thresholds.

A primary challenge in using existing methods in our application is that they cannot discern manipulation from truthful and legal rounding. Rounding to the same decimal as the regulatory thresholds is indeed allowed under the LCR and is an important data feature. We show that existing methods find pervasive manipulation around the regulatory thresholds before and after 2016 despite heightened scrutiny around lead monitoring following the public revelation of the Flint water crisis. Applying our method, we find strong evidence of excess rounding between 2011 and 2015 but no statistically significant excess rounding between 2016 and 2020. The findings highlight an important empirical setting where existing methods suggest prolonged manipulation, even after heightened scrutiny around lead monitoring, whereas our methods do not.

We contribute to several branches of the economics literature. We expand the existing literature on threshold manipulation, which relies on discontinuity-based and/or bunching estimators to detect manipulation. The literature has documented a range of settings where agents manipulate data. Examples include death and population statistics that determine officials’ promotions and grant allocations (Fisman and Wang 2017; Foremny, Jofre-Monseny, and Solé-Ollé 2017), high-stakes exam scores that determine educational funding (Jacob 2005; Figlio 2006; Cullen and Reback 2006; Diamond and Persson 2016; Dee et al. 2019), index scores that determine social program eligibility (Camacho and Conover 2011), and environmental

²Our methods apply in settings where rounding is allowed but not required under the policy rule.

statistics that determine regulatory status or government officials' promotions (Sneeringer and Key 2011; Ghanem and Zhang 2014; Grainger and Schreiber 2019; Zou 2021).

Related work has documented instances where agents manipulate data and rounding occurs. For example, rounding is prominent in public procurement auctions data (Palguta and Pertold 2017), tax reporting data (Kleven and Waseem 2013), birth weight data (Barreca et al. 2011), and birth dates and annual income data (Barreca, Lindo, and Waddell 2016). Where authors have acknowledged rounding, they have explored the sensitivity of their preferred estimators to more flexible specifications, investigated specifications using only rounded or nonrounded data, or attempted to control for it in a regression framework, for example, by including round number fixed effects. We, in contrast, formally model rounding behavior and demonstrate limitations of existing manipulation measures even when a small percentage of data are rounded to the policy threshold.

We also contribute to a robust literature on the economic impacts, health effects, and regulation of lead (Reyes 2007; Ferrie, Rolf, and Troesken 2012; Clay, Troesken, and Haines 2014; Reyes 2015; Feigenbaum and Muller 2016; Lanphear et al. 2018; Aizer and Currie 2019; Grossman and Slusky 2019; Hollingsworth and Rudik 2021; Christensen, Keiser, and Lade 2023). We identify issues in the implementation of the LCR that may compromise its ability to protect public health. Finally, we add to a recent literature on reporting behavior under major US environmental statutes (Bennear, Jessoe, and Olmstead 2009; Grainger, Schreiber, and Chang 2019; Grainger and Schreiber 2019; Zou 2021; Ding et al. 2022). We highlight a situation where increased scrutiny decreased suspicious behavior and spurred better adherence to regulatory guidance.

I. Testing and Quantifying Threshold Manipulation in the Presence of Rounding

Consider a potential outcomes framework that allows for both manipulation and rounding.³ Throughout the paper, and without loss of generality, we assume that the direction of manipulation is to move observations below policy thresholds. However, our results extend to settings where threshold manipulation takes the opposite direction. Let X denote the observed policy variable. Define Z as an indicator for whether X is manipulated, $X(0)$ the unmanipulated variable, and $X(1)$ the manipulated variable. Thus, the observed variable X can be written as

$$X = (1 - Z)X(0) + ZX(1).$$

Suppose further that agents may report rounded data. We allow for rounding absent manipulation by introducing an indicator, R , that equals one when a value is rounded. This allows us to write $X(0)$ as

$$X(0) = RX^R(0) + (1 - R)X^N(0),$$

³We follow the literature and refer to behavior that alters the true value of an official statistic as manipulation. As in the literature, manipulation refers to a broad class of underlying behaviors, including intentional misreporting and reporting errors.

where $X^R(0)$ and $X^N(0)$ denote the unmanipulated variable with and without rounding, respectively.

Existing approaches for testing and quantifying threshold manipulation (e.g., McCrary 2008; Ghanem, Shen, and Zhang 2020) assume continuity absent manipulation. Since rounding may lead to discontinuities absent manipulation, applying existing approaches in those settings would rely on a counterfactual absent manipulation *and* rounding, formally $X^N(0)$, and thereby confound the two behaviors. As a result, manipulation tests or measures that assume continuity absent manipulation may lead to biased conclusions in the presence of rounding.

To demonstrate the potential bias of existing methods in the presence of rounding, consider the proportion of overall threshold manipulation $\lambda_j = F_X(c_j) - F_{X^N(0)}(c_j)$ proposed in Ghanem, Shen, and Zhang (2020). Let $F_W(\cdot)$ denote the cumulative distribution function (CDF) for a random variable W and $\{c_j\}_{j=1}^J$ the thresholds around which agents may manipulate data. Using our potential outcomes framework and adding and subtracting $F_{X(0)}(c_j)$, we can decompose this threshold manipulation measure into two components:

$$\lambda_j = \underbrace{F_X(c_j) - F_{X(0)}(c_j)}_{\text{Proportion of threshold manipulation}} + \underbrace{F_{X(0)}(c_j) - F_{X^N(0)}(c_j)}_{\text{Rounding bias}}.$$

The first term, the difference between the observed and unmanipulated distributions at c_j , is the proportion of threshold manipulation. The second term is the proportion of rounding, which is the difference between the distribution of $X(0)$ and $X^N(0)$ at c_j . If $X^N(0)$ is rounded down to c_j , a concern in our setting, this moves mass to c_j in the distribution of $X(0)$, formally $F_{X(0)}(c_j) > F_{X^N(0)}(c_j)$. It follows that rounding bias is positive, and λ_j will be an upwardly biased measure of threshold manipulation. Note that if agents only round up to c_j , then there would be no rounding bias.⁴

To overcome the shortcomings of existing methods, we introduce a new hypothesis test for threshold manipulation (Section IA) and a new manipulation measure (Section IB). Both rely on comparisons of the observed and counterfactual distribution absent threshold manipulation but accounting for rounding.

A. Manipulation Testing in the Presence of Rounding

Define a “roundable interval” $[l_j, u_j)$ around each c_j as an interval that contains all values that may be rounded to c_j .⁵ Our manipulation test relies on the key insight that rounding only moves mass within these roundable intervals. As a result, absent manipulation, the proportion of values within a roundable interval in the observed distribution of X should equal the respective proportion in the unmanipulated distribution of $X(0)$. This insight yields the following implication of no threshold manipulation around c_j for $j = 1, \dots, J < \infty$:

$$(1) \quad \Pr(Z = 1 | X \in (\underline{x}_j, \bar{x}_j)) = 0 \Rightarrow \Pr(l_j \leq X < u_j) = \Pr(l_j \leq X(0) < u_j),$$

⁴ See Supplemental Appendix A.1 for a discussion of rounding bias when manipulation is upward.

⁵ Defining roundable intervals is, naturally, context-specific.

where $(\underline{x}_j, \bar{x}_j)$ denotes the manipulation window around c_j such that threshold manipulation can lead values in $(c_j, \bar{x}_j)$ to fall inside $(\underline{x}_j, c_j]$.⁶ The left-hand side of the implication in equation (1) is the observed probability of being within a roundable interval around c_j . The right-hand side is unobserved. As a result, to test this implication, it remains to identify $\Pr(l_j \leq X(0) < u_j)$.

Building further on this insight, it follows that the probability of being in the roundable interval around the policy cutoff c_j is equal for $X(0)$ and $X^N(0)$. To see this, it is helpful to formalize the assumption that only values in the roundable interval can be rounded to c_j .

ASSUMPTION 1 (Roundable Interval for Policy Cutoff): $\Pr(X^R(0) = c_j, R = 1) = \Pr(l_j \leq X^N(0) < u_j, R = 1)$ for $j = 1, \dots, J$.

Assumption 1 requires the knowledge of the roundable interval and does not allow for rounding absent manipulation to lead to c_j if $X^N(0) \notin [l_j, u_j]$. Within the roundable interval, rounding can be up and/or down to c_j .

Under Assumption 1, we can show that the probabilities of $X(0)$ and $X^N(0)$ being in the roundable interval are equal.

$$\begin{aligned}
 (2) \quad \Pr(l_j \leq X(0) < u_j) &= \Pr(l_j \leq X(0) < u_j, R = 1) \\
 &\quad + \Pr(l_j \leq X(0) < u_j, R = 0) \\
 &= \Pr(X^R(0) = c_j, R = 1) + \Pr(l_j \leq X(0) < u_j, R = 0) \\
 &= \Pr(l_j \leq X^N(0) < u_j, R = 1) \\
 &\quad + \Pr(l_j \leq X^N(0) < u_j, R = 0) \\
 &= \Pr(l_j \leq X^N(0) < u_j).
 \end{aligned}$$

The first and last equalities follow from the law of total probability. The second equality follows from the definition of $X(0)$, which implies that when $R = 1$, $X(0) = X^R(0)$, as well as the definition of the roundable interval. The third equality follows from Assumption 1 and $X(0) = X^N(0)$ when $R = 0$.

Existing approaches identify the distribution of $X^N(0)$ under suitable conditions. We rely on the censored maximum likelihood approach in Ghanem, Shen, and Zhang (2020). To adapt their approach to our setting, we impose their assumptions on $X^N(0)$ (the unrounded, unmanipulated variable) as opposed to $X(0)$. We formalize these assumptions and use $\mathbb{X}$ to denote the support of $X^N(0)$.

⁶Note that to differentiate between rounding and threshold manipulation in this context, $(c_j, u_j) \subset (c_j, \bar{x}_j)$.

ASSUMPTION 2 (Adapted from Ghanem, Shen, and Zhang 2020): Let $\{(\underline{x}_j, \bar{x}_j)\}_{j=1}^J$ be a set of nonoverlapping manipulation windows around policy thresholds $\{c_j\}_{j=1}^J$, respectively, where $\underline{x}_1 < c_1 < \bar{x}_1 < \underline{x}_2 < c_2 < \bar{x}_2 < \dots < \underline{x}_J < c_J < \bar{x}_J$. The following assumptions hold:

1. $\Pr(Z = 1 | X^N(0) = x) = 0$ for $x \in \mathbb{X} \setminus \bigcup_{j=1}^J (\underline{x}_j, \bar{x}_j)$.
2. For $j = 1, \dots, J$, $X(1) \in (\underline{x}_j, c_j]$ if $X^N(0) \in (c_j, \bar{x}_j)$; $Z = 0$ if $X^N(0) \in \bigcup_{j=1}^J (\underline{x}_j, c_j]$.
3. $F_{X^N(0)}(\cdot) = G(\cdot; \theta)$, where $G(x; \theta)$ is continuously differentiable in $x \in \text{int}(\mathbb{X})$ with density $g(x; \theta)$. $G(\cdot; \theta)$ and $g(\cdot; \theta)$ are known functions up to an unknown finite-dimensional parameter θ .

Assumption 2 consists of three conditions. Assumption 2.1 implies that outside the manipulation windows we observe the unmanipulated variable, such that $X = X(0)$.⁷ Assumption 2.2 requires that manipulation occurs in a single, known direction within the manipulation window, moving a value across the policy threshold. Motivated by our application, we assume that manipulation around a given policy cutoff c_j occurs when $X^N(0)$ is above the policy cutoff c_j and is manipulated to a value $X(1)$ that is below c_j . Finally, Assumption 2.3 requires the distribution of $X^N(0)$ to belong to a parametric class of distributions, $G(\cdot; \theta)$. Following Ghanem, Shen, and Zhang (2020), we specify $G(\cdot; \theta)$ as the generalized beta distribution of the second kind (GB2). This class of distributions was introduced in McDonald (1984) as a flexible class of distributions that is particularly well suited for income distributions as it includes the lognormal, gamma, and Weibull distributions. The GB2 class also nests several distributions that fit pollutant concentrations well (Ghanem, Shen, and Zhang 2020).

Using Assumptions 1 and 2, we can rewrite equation (2) as follows:

$$(3) \quad \Pr(l_j \leq X(0) < u_j) = F_{X^N(0)}(u_j) - F_{X^N(0)}(l_j) = G(u_j; \theta) - G(l_j; \theta).$$

Plugging the right-hand side of equation (3) into (1) yields the following testable implication of no threshold manipulation.⁸

PROPOSITION 1 (Testable Implication of No Threshold Manipulation in the Presence of Rounding): Suppose that Assumptions 1 and 2 hold. Furthermore, $(c_j, u_j) \subset (c_j, \bar{x}_j)$ for $j \in \{1, \dots, J\}$ and $l_j > \bar{x}_{j-1}$ for $j \in \{2, \dots, J\}$. Then, for $j \in \{1, \dots, J\}$, if $\Pr(Z = 1 | X \in (\underline{x}_j, \bar{x}_j)) = 0$, then $\Pr(l_j \leq X < u_j) = G(u_j; \theta) - G(l_j; \theta)$.

⁷If additional types of manipulation change other aspects of the distribution, then the distribution of $X(0)$ should be interpreted as being absent threshold manipulation but exhibiting these other forms of manipulation.

⁸We provide a more detailed proof of Proposition 1 in Supplemental Appendix A.2.

Proposition 1 yields the following null and alternative hypotheses of our manipulation test for $j = 1, \dots, J$:

$$(4) \quad \begin{aligned} H_0^j: \Pr(l_j \leq X < u_j) &= G(u_j; \theta) - G(l_j; \theta), \\ H_1^j: \Pr(l_j \leq X < u_j) &\neq G(u_j; \theta) - G(l_j; \theta). \end{aligned}$$

Our manipulation test relies on a two-sided alternative since manipulation can add mass into the roundable interval or reduce it. Therefore, $\Pr(l_j \leq X < u_j) - \Pr(l_j \leq X(0) < u_j)$ could be positive or negative. For example, manipulating values that are greater than u_j to fall within $[l_j, u_j)$ would lead to $\Pr(l_j \leq X < u_j) - \Pr(l_j \leq X(0) < u_j) > 0$, while manipulating values in (c_j, u_j) to be less than l_j would lead to $\Pr(l_j \leq X < u_j) - \Pr(l_j \leq X(0) < u_j) < 0$.

A distinctive feature of our manipulation test, relative to our excess rounding measure, is that it does not require any additional assumptions on rounding behavior except the knowledge of the roundable interval (Assumption 1). However, a shortcoming of our test is that its statistic is not monotonic in the extent of manipulation, as we described earlier, and should not be interpreted as a magnitude of manipulation.⁹ We, therefore, examine how to quantify manipulation in the presence of rounding.

B. Identifying Excess Rounding

Here, we introduce a new threshold manipulation measure called “excess rounding.” For each threshold c_j , we define excess rounding as the difference between the observed and counterfactual probabilities of rounding at the threshold:

$$(5) \quad \eta_j = \Pr(X = c_j) - \Pr(X(0) = c_j).$$

The measure quantifies data manipulation that leads to reported values exactly equal to the relevant policy thresholds. Identifying η_j requires estimating two objects. The proportion of rounding at threshold c_j is directly observed from the data as

$$(6) \quad \Pr(X = c_j) = \lim_{x \downarrow c_j} F_X(x) - \lim_{x \uparrow c_j} F_X(x).$$

The counterfactual proportion of rounding at threshold c_j , $\Pr(X(0) = c_j)$ can be decomposed into two components:

$$(7) \quad \Pr(X(0) = c_j) = \Pr(X(0) = c_j | l_j \leq X(0) < u_j) \Pr(l_j \leq X(0) < u_j).$$

The first term is the counterfactual probability of rounding at the policy threshold conditional on being within the roundable interval. The second term is the counterfactual probability of being within the roundable interval, which we also rely on

⁹The McCrary test statistic also suffers from this issue.

in our hypothesis test and is identified in equation (3) under Assumptions 1 and 2. Plugging equation (3) into (7) yields

$$(8) \quad \Pr(X(0) = c_j) = \Pr(X(0) = c_j | l_j \leq X(0) < u_j) [G(u_j; \theta) - G(l_j; \theta)].$$

To identify the excess rounding measure, it remains to identify the counterfactual probability of rounding to the policy threshold absent manipulation and conditional on being in the roundable interval $\Pr(X(0) = c_j | l_j \leq X(0) < u_j)$. Identifying this probability requires additional assumptions. We consider two approaches, both exploiting rounding behavior away from the policy thresholds. The first yields a point identification result. The second bounds the probability.

Our first approach requires the existence of a set of nonpolicy thresholds, $\mathcal{R}_j$, in the support of $X(0)$, where the probability of rounding (absent manipulation) conditional on being within the respective roundable intervals equals the conditional probability of rounding around the policy thresholds c_j absent manipulation. For example, in our application, rounding occurs at the thousandth decimal. Thus, natural nonpolicy thresholds for $\mathcal{R}_j$ are the thousandth decimals of X outside the manipulation windows.¹⁰ Under Assumption 3, we assume that rounding away from the policy threshold is homogeneous.

ASSUMPTION 3 (Homogeneous Rounding Absent Manipulation): *For $j = 1, \dots, J$, the following assumptions hold for $r \in \mathcal{R}_j$:*

1. $\Pr(R = 1, X(0) = r) = \Pr(R = 1, l_r \leq X^N(0) < u_r) \quad \text{and} \quad [l_r, u_r) \cap \bigcup_{j=1}^J (x_j, \bar{x}_j) = \emptyset.$
2. $\Pr(X(0) = c_j | l_j \leq X(0) < u_j) = \Pr(X = r | l_r \leq X < u_r).$

Assumption 3.1 rules out any overlap between the roundable intervals for $r \in \mathcal{R}_j$ and the manipulation windows, which implies that the probability mass at r is only due to rounding and not manipulation. That is, $\Pr(X = r | l_r \leq X < u_r) = \Pr(X(0) = r | l_r \leq X(0) < u_r)$. Assumption 3.2 ensures that for each policy threshold c_j , there is a set of values $\mathcal{R}_j$, where the counterfactual conditional probability of rounding absent manipulation, $\Pr(X(0) = c_j | l_j \leq X(0) < u_j)$, equals the observed conditional probability of rounding in those intervals, $\Pr(X = r | l_r \leq X < u_r)$ for $r \in \mathcal{R}_j$.¹¹

Under Assumption 3, the (unconditional) probability of rounding absent manipulation at policy threshold c_j in equation (8) is given by

$$(9) \quad \Pr(X(0) = c_j) = \Pr(X(0) = c_j | l_j \leq X(0) < u_j) [G(u_j; \theta) - G(l_j; \theta)] \\ = \frac{1}{|\mathcal{R}_j|} \sum_{r \in \mathcal{R}_j} \Pr(X = r | l_r \leq X < u_r) [G(u_j; \theta) - G(l_j; \theta)],$$

¹⁰The choice of $\mathcal{R}_j$ is context-specific but should reflect nonpolicy cutoffs that best mimic rounding at the policy threshold absent threshold manipulation.

¹¹Note that if R is independent of $X^N(0)$, then $\Pr(R = 1 | l_r \leq X(0) < u_r) = \Pr(R = 1)$, which yields homogeneous rounding at all rounding cutoffs, absent manipulation.

where we take a simple average of the conditional probability of rounding over $r \in \mathcal{R}_j$ and $|\mathcal{R}_j|$ denotes the number of elements in the set $\mathcal{R}_j$. As a result, the proportion of excess rounding around policy threshold c_j for $j = 1, \dots, J$ is point identified. In the following proposition, we provide the resulting identification result for η_j , but instead of using the simple average of the conditional probabilities, we allow for any user-specified convex combination using weights $\{\omega_r\}_{r \in \mathcal{R}_j}$.

PROPOSITION 2 (Point Identification of Excess Rounding): *Suppose that Assumptions 1–3 hold. Then, for $j = 1, \dots, J$,*

$$(10) \quad \eta_j^\omega = \Pr(X = c_j) - \sum_{r \in \mathcal{R}_j} \omega_r \Pr(X = r | l_r \leq X < u_r) [G(u_j; \theta) - G(l_j; \theta)],$$

where $\omega_r \geq 0$ for $r \in \mathcal{R}_j$ and $\sum_{r \in \mathcal{R}_j} \omega_r = 1$.

The proof of Proposition 2 is immediate from equations (5) and (9). Under Assumption 3, one can use $\Pr(X = r | l_r \leq X < u_r)$ for any $r \in \mathcal{R}_j$ to identify $\Pr(X(0) = c_j | l_j \leq X(0) < u_j)$. As a result, any convex combination of $\{\Pr(X = r | l_r \leq X < u_r)\}_{r \in \mathcal{R}_j}$ would be valid. In equation (9), we set $\omega_r = 1/|\mathcal{R}_j|$ as it is the simplest from an illustration perspective, which yields the following estimand:

$$\eta_j^s = \Pr(X = c_j) - \frac{1}{|\mathcal{R}_j|} \sum_{r \in \mathcal{R}_j} \Pr(X = r | l_r \leq X < u_r) [G(u_j; \theta) - G(l_j; \theta)].$$

The probability-weighted average, which sets $\omega_r = \frac{\Pr(l_r \leq X < u_r)}{\sum_{r \in \mathcal{R}_j} \Pr(l_r \leq X < u_r)}$, admits the following convenient simplification, which we derive in Supplemental Appendix A.3:

$$(11) \quad \eta_j^p = \Pr(X = c_j) - \Pr\left(R = 1 \middle| X \in \bigcup_{r \in \mathcal{R}_j} [l_r, u_r)\right) [G(u_j; \theta) - G(l_j; \theta)].$$

Since homogeneous rounding may be an overly restrictive assumption in some settings, we also consider bounds on η_j under a bounded rounding condition. Under this assumption, we argue that rounding at nonpolicy thresholds provides extrema for rounding at the policy threshold.

ASSUMPTION 4 (Bounded Rounding Absent Manipulation): *For $j = 1, \dots, J$, the following assumptions hold:*

1. $\Pr(R = 1, X(0) = r) = \Pr(R = 1, l_r \leq X^N(0) < u_r), [l_r, u_r) \cap \bigcup_{j=1}^J (\underline{x}_j, \bar{x}_j) = \emptyset$ for $r \in \mathcal{R}_j$.
2. $\min_{r \in \mathcal{R}_j} \Pr(X = r | l_r \leq X < u_r) \leq \Pr(X(0) = c_j | l_j \leq X(0) < u_j) \leq \max_{r \in \mathcal{R}_j} \Pr(X = r | l_r \leq X < u_r)$.

Assumption 4.1 is identical to Assumption 3.1, which ensures that there is no manipulation in the roundable interval around r for $r \in \mathcal{R}_j$. Assumption 4.2 requires the conditional probability of rounding at the policy threshold absent manipulation to lie between the minimum and maximum of the observed conditional probability of rounding over the set of values in $\mathcal{R}_j$. Under Assumption 4, we can bound $\Pr(X(0) = c_j)$ as follows:

$$\begin{aligned}
 (12) \quad & \min_{r \in \mathcal{R}_j} \Pr(X = r | l_r \leq X < u_r) \Pr(l_j \leq X(0) < u_j) \\
 & \leq \Pr(X(0) = c_j) \\
 & \leq \max_{r \in \mathcal{R}_j} \Pr(X = r | l_r \leq X < u_r) \Pr(l_j \leq X(0) < u_j),
 \end{aligned}$$

which simplifies under Assumption 2 to

$$\begin{aligned}
 (13) \quad & \min_{r \in \mathcal{R}_j} \Pr(X = r | l_r \leq X < u_r) [G(u_j; \boldsymbol{\theta}) - G(l_j; \boldsymbol{\theta})] \\
 & \leq \Pr(X(0) = c_j) \\
 & \leq \max_{r \in \mathcal{R}_j} \Pr(X = r | l_r \leq X < u_r) [G(u_j; \boldsymbol{\theta}) - G(l_j; \boldsymbol{\theta})].
 \end{aligned}$$

The following proposition provides the resulting bounds on η_j , which follow from equations (5) and (13).

PROPOSITION 3: *Suppose that Assumptions 1, 2, and 4 hold. Then,*

$$\begin{aligned}
 (14) \quad \eta_j \in & \left[\Pr(X = c_j) - \max_{r \in \mathcal{R}_j} \Pr(X = r | l_r \leq X < u_r) [G(u_j; \boldsymbol{\theta}) - G(l_j; \boldsymbol{\theta})], \right. \\
 & \left. \Pr(X = c_j) - \min_{r \in \mathcal{R}_j} \Pr(X = r | l_r \leq X < u_r) [G(u_j; \boldsymbol{\theta}) - G(l_j; \boldsymbol{\theta})] \right].
 \end{aligned}$$

C. Estimation

We estimate $\boldsymbol{\theta}$ via censored maximum likelihood estimation (CMLE) using the following log-likelihood (Ghanem, Shen, and Zhang 2020) to approximate the counterfactual distribution $G(\cdot; \boldsymbol{\theta})$:

$$\begin{aligned}
 (15) \quad \mathcal{L} = & \sum_{i=1}^n \left\{ \mathbf{1} \left\{ X_i \notin \bigcup_{j=1}^J (x_j, \bar{x}_j) \right\} \log g(X_i; \boldsymbol{\theta}) \right. \\
 & \left. + \sum_{j=1}^J \mathbf{1} \{ X_i \in (x_j, \bar{x}_j) \} \log (G(\bar{x}_j; \boldsymbol{\theta}) - G(x_j; \boldsymbol{\theta})) \right\},
 \end{aligned}$$

where $g(\cdot; \boldsymbol{\theta})$ and $G(\cdot; \boldsymbol{\theta})$ are the probability density function (PDF) and CDF, respectively, of the GB2 distribution.

To test the null hypothesis in equation (4), we use the test statistic

$$(16) \quad T^j = \left| \frac{\hat{\Delta}_j}{\text{SE}(\hat{\Delta}_j)} \right|, \text{ where } \hat{\Delta}_j = \widehat{\Pr}(l_j \leq X < u_j) - [G(u_j; \hat{\theta}) - G(l_j; \hat{\theta})],$$

where for an event A , $\widehat{\Pr}(A) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{A_i\}$. We compute the standard error ($\text{SE}(\hat{\Delta}_j)$) using the nonparametric bootstrap, which is valid under Assumptions 1 and 2 as well as additional regularity conditions that ensure the joint asymptotic normality of the maximum likelihood estimator, $\hat{\theta}$, and $\widehat{\Pr}(l_j \leq X < u_j)$.

We estimate the excess rounding measure under Assumption 3 by plugging in sample analogs in place of population quantities into equation (10) using the simple and probability-weighted averages as follows:

$$(17) \quad \hat{\eta}_j^s = \widehat{\Pr}(X = c_j) - \frac{1}{|\mathcal{R}_j|} \sum_{r \in \mathcal{R}_j} \widehat{\Pr}(X = r | l_r \leq X < u_r) [G(u_j; \hat{\theta}) - G(l_j; \hat{\theta})],$$

$$\hat{\eta}_j^p = \widehat{\Pr}(X = c_j) - \widehat{\Pr}\left(R = 1 \mid X \in \bigcup_{r \in \mathcal{R}_j} [l_r, u_r)\right) [G(u_j; \hat{\theta}) - G(l_j; \hat{\theta})].$$

The estimators $\hat{\eta}_j^s$ and $\hat{\eta}_j^p$ are consistent for η_j under Assumptions 1–3 as well as suitable regularity conditions. Their asymptotic normality follows from the delta method under additional regularity conditions that ensure the joint normality of the relevant sample probabilities and the maximum likelihood estimator $\hat{\theta}$.¹² The standard errors can, therefore, be consistently estimated via nonparametric bootstrap.

Similarly, $\hat{\eta}_j^{LB}$ and $\hat{\eta}_j^{UB}$ denote the estimates of the lower and upper bounds on η_j in equation (14), respectively, and are given by

$$(18) \quad \hat{\eta}_j^{LB} = \widehat{\Pr}(X = c_j) - \max_{r \in \mathcal{R}_j} \widehat{\Pr}(X = r | l_r \leq X_i < u_r) [G(u_j; \hat{\theta}) - G(l_j; \hat{\theta})],$$

$$\hat{\eta}_j^{UB} = \widehat{\Pr}(X = c_j) - \min_{r \in \mathcal{R}_j} \widehat{\Pr}(X = r | l_r \leq X < u_r) [G(u_j; \hat{\theta}) - G(l_j; \hat{\theta})].$$

Since the bounds include nondifferentiable functions of estimated quantities, the delta method does not apply, and inference is nonstandard and beyond the scope of the present paper. The upper and lower bound estimates are useful nonetheless to check the sensitivity of the magnitude of $\hat{\eta}_j$ to deviations from the homogeneous rounding assumption.

D. Monte Carlo Simulations

In this section, we compare the finite-sample performance of our proposed test and manipulation measure with existing approaches using an empirically calibrated

¹²Sufficient conditions for the asymptotic normality of $\hat{\eta}_j^s$ ($\hat{\eta}_j^p$) require that $\Pr(X = c_j)$ and $\frac{1}{|\mathcal{R}_j|} \sum_{r \in \mathcal{R}_j} \Pr(X = r | l_r \leq X_j < u_r) (\Pr(R = 1 | X \in \bigcup_{r \in \mathcal{R}_j} [l_r, u_r)))$ are bounded away from zero. While one could alternatively use an efficient generalized method of moments approach using the sample moments for $r \in \mathcal{R}_j$, $\hat{\eta}_j^s = \widehat{\Pr}(X = c_j) - \widehat{\Pr}(X = r | l_r \leq X < u_r) [G(u_j; \hat{\theta}) - G(l_j; \hat{\theta})]$, this approach would require $\Pr(X = r | l_r \leq X < u_r)$ to be bounded away from zero for each $r \in \mathcal{R}_j$, which is stronger than the relevant condition for $\hat{\eta}_j^s$ and $\hat{\eta}_j^p$. A comparison of these different estimation approaches is left for future work.

TABLE 1—MONTE CARLO SIMULATION DESIGN AND RESULTS

Panel A. Design

Unmanipulated variable absent rounding:

$X^N(0)$ is generated using GB2 class with parameter θ

Rounding indicator:

$R \overset{iid}{\sim} \text{Bernoulli}(p_R)$

Unmanipulated variable:

$X(0) = RX^R(0) + (1 - R)X^N(0)$

Manipulation indicator:

$Z \overset{iid}{\sim} \text{Bernoulli}(p_Z)$

Observed variable:

$$X = \begin{cases} X(0), & \text{if } X^N(0) \notin (c, \bar{x}) \\ (1 - Z)X(0) + Zc, & \text{if } X^N(0) \in (c, \bar{x}) \end{cases}$$

Panel B. Results

p_R (%)	$P(X(0) = c)$ (%)	McCrary test	AGKL test	Overall manipulation ($\hat{\lambda}\%$)			Excess rounding					
							$\hat{\eta}^s(\%)$			$\hat{\eta}^p(\%)$		
				$\hat{p}_\alpha(\%)$	$\hat{p}_\alpha(\%)$	Mean	SD	SE/SD	Mean	SD	SE/SD	Mean
0% threshold manipulation at c												
0	0.00	4.8	6.2	0.01	0.24	0.98	0.00	0.00	—	0.00	0.00	—
5	0.39	73.3	4.7	0.18	0.23	1.01	0.08	0.06	0.99	0.01	0.05	1.01
10	0.77	100.0	4.8	0.36	0.23	1.00	0.15	0.08	0.99	0.02	0.07	1.00
2.5% threshold manipulation at c												
0	0.00	100.0	100.0	2.49	0.23	1.00	2.50	0.13	0.98	2.50	0.13	0.98
5	0.39	100.0	100.0	2.66	0.23	1.00	2.55	0.14	0.98	2.48	0.14	0.97
10	0.77	100.0	100.0	2.80	0.23	0.99	2.59	0.15	0.99	2.46	0.15	0.99
5% threshold manipulation at c												
0	0.00	100.0	100.0	4.99	0.23	0.98	5.00	0.18	1.00	5.00	0.18	1.00
5	0.39	100.0	100.0	5.12	0.22	1.00	5.01	0.19	0.98	4.95	0.19	0.98
10	0.77	100.0	100.0	5.23	0.22	1.00	5.02	0.19	0.98	4.89	0.19	0.98

Notes: Panel A presents our simulation design. We calibrate it to our empirical application in multiple ways. First, the unmanipulated variable absent rounding ($X^N(0)$) follows the GB2 distribution with parameters set to the estimates from the GB2 fit for the lead monitoring data from medium water systems in our application. Second, we allow for rounding to three decimal points. Third, we use $c = 0.005$ as the policy threshold. Finally, we set $n = 15,000$, the approximate sample size for medium water systems pre-Flint. Panel B presents the simulation results across 1,000 simulation replications. In our design, $p_R = \Pr(X(0) = r | l_r \leq X^N(0) < u_r)$ for all r , and the proportion of rounding to c is given by $\Pr(X(0) = c) = p_R[G(u; \theta) - G(l; \theta)]$. The proportion of threshold manipulation is given by $\Pr(Z = 1, X^N(0) \in (c, \bar{x})) = p_Z[G(\bar{x}; \theta) - G(c; \theta)]$. We therefore set $p_Z = \Pr(Z = 1, X^N(0) \in (c, \bar{x})) / [G(\bar{x}; \theta) - G(c; \theta)]$ to target a specific proportion of threshold manipulation. For the McCrary test and our proposed test (“AGKL test”), we report $\hat{p}_\alpha$, the simulation rejection probability using nominal level $\alpha = 5\%$. For $\hat{\lambda}$, $\hat{\eta}^s$, and $\hat{\eta}^p$, we report the simulation mean, the simulation standard deviation (SD), and the ratio of the simulation mean of the standard error and the simulation standard deviation (SE/SD). We estimate the standard error using nonparametric bootstrap with 200 replications and set the manipulation window to (0.0035, 0.0075) in the censored maximum likelihood objective function.

simulation study with various combinations of probabilities of rounding and threshold manipulation. Panel A of Table 1 presents our simulation design.¹³ First, we generate the unmanipulated variable absent manipulation, $X^N(0)$, from a GB2 distribution with a sample size of 15,000. Second, we generate the rounding indicator, R , as a Bernoulli random variable where p_R is 0 percent, 5 percent, or 10 percent.¹⁴ Motivated by our empirical application, we use this indicator to allow for rounding to the thousandth decimal place away from zero. Last, to

¹³The sample size and GB2 parameter choices follow the sample size and estimates of the medium water systems during the pre-Flint period in our application.

¹⁴To satisfy Assumption 3, we assume that rounding (R) is independent of $X^N(0)$.

simplify illustration, we allow for threshold manipulation around one policy cutoff, $c = 0.005$. We generate the manipulation indicator, Z , as a Bernoulli random variable and consider different degrees of threshold manipulation at c : 0 percent, 2.5 percent, and 5 percent.

The simulation results in panel B of Table 1 demonstrate the main takeaways from our theoretical analysis. In the absence of threshold manipulation, even for very small proportions of rounding at c , the McCrary test falsely rejects the null of no threshold manipulation across the majority of simulation replications (73 percent or higher), whereas our test controls size when there is no threshold manipulation, regardless of the presence of rounding. Furthermore, our test demonstrates nontrivial power in finite samples when there is threshold manipulation. For the case where there is neither threshold manipulation nor rounding, both the McCrary test and our test control size.¹⁵

Next, we compare $\hat{\lambda}$, $\hat{\eta}^s$, and $\hat{\eta}^p$ in the simulation results. Since manipulation leads to values exactly equal to the cutoff in our design, specifically $X = c$ if $Z = 1$, the excess rounding proportion η equals the probability of threshold manipulation. As a result, all estimators are consistent for the true proportion of threshold manipulation in the absence of rounding. In the presence of rounding, however, we note that $\hat{\lambda}$ tends to have an upward bias, whereas the means of $\hat{\eta}^s$ and $\hat{\eta}^p$ are closer to the true proportion of threshold manipulation. Finally, the simulation results demonstrate that the nonparametric bootstrap leads to standard errors with good finite-sample properties, as the ratio of the simulation mean of the standard error and the simulation standard deviation is close to one across all designs.

II. Threshold Manipulation in US Lead Monitoring

We apply our method to test for and quantify threshold manipulation in lead reporting required by the LCR, a federal regulation established by the EPA under the authority of the US Safe Drinking Water Act and the primary regulation aimed at reducing US residents' exposure to lead in drinking water. The rule sets monitoring and remediation requirements for US PWSs, covering upward of 90 percent of the US population (USEPA 2022). The rule requires that water systems periodically collect lead samples from high-risk homes in their service territories. PWSs report the ninetieth percentile of the sample values for lead. If the value exceeds a lower "reporting" threshold, 0.005 milligrams per liter (mg/L), systems may need to continue conducting frequent monitoring. If the values exceed a higher "action" level, 0.015 mg/L, systems must take costly actions to reduce lead levels and educate the public.¹⁶ Exceeding either threshold increases regulatory scrutiny and anticipated costs, potentially incentivizing data manipulation.

Three features of the LCR motivate our study of potential threshold manipulation. First, the rule provides several avenues through which PWSs can manipulate their reported ninetieth-percentile lead values (Roy and Edwards 2019; Giles 2020).

¹⁵We drop values greater than 0.02 when computing the McCrary test in the simulation study as it has a tendency to over-reject, even in the absence of rounding and manipulation, when it is estimated for the full sample. The proportion of dropped observations is negligible.

¹⁶While the rule also requires copper monitoring, we focus exclusively on lead because copper ninetieth-percentile values are only available for systems exceeding the copper action level of 1.3 mg/L.

These “cheats” include sampling lower-risk homes, using collection bottles with small openings, altering the pre-stagnation period, removing aerators before sampling, and taking additional samples beyond the minimum in the rule from lower-risk homes, a practice known as “sampling out.” Second, the LCR reporting guidance allows PWSs to round their data to the thousandth decimal, the same decimal as the policy thresholds. As a result, rounding is common in the reported data. Third, scrutiny around reported lead concentrations increased dramatically following the Flint water crisis, allowing us to examine whether reporting behavior changed between two periods with distinctly different levels of regulatory scrutiny.

A. Data and Summary Statistics

We obtain water system-level data for 2011 to 2020 from the US EPA’s Enforcement and Compliance History Online database (USEPA 2021). For each PWS, we extract data on the number of people it serves, the system type, the dates of the monitoring period, and the reported ninetieth-percentile lead values during that monitoring period. We restrict our sample to active water systems, drop observations equal to zero or nondetects to avoid bottom-censoring issues, and drop lead concentration values that exceed 100 mg/L as these extremely high values are likely reporting errors.

We split the data along two dimensions. First, we split the data by system size since LCR prescriptions differ depending on whether a system is small, medium, or large.¹⁷ We further divide the data into pre- versus post-January 2016, which we call pre- versus post-Flint, for two reasons. First, awareness about the Flint water crisis increased substantially in January 2016 when a federal state of emergency was declared. Second, in response to the Flint water crisis, the EPA issued lead monitoring recommendations in 2016 to “better reflect the state of knowledge about the fate and transport of lead in distribution systems” (Grevatt 2016). These recommendations signaled greater federal scrutiny over lead reporting and explicitly discouraged using many of the “cheats” described above.¹⁸

Figure 1 presents a data histogram. Two features are noteworthy. First, rounding to the thousandth decimal place is common. Second, rounding decreased post-Flint, particularly at the reporting threshold.¹⁹ The change is most evident for medium PWSs.

B. Results

Before presenting our primary results, we highlight the limitations of existing methods in our setting. We focus on ninetieth-percentile lead reports by medium-sized systems at the reporting threshold since they saw the largest changes in rounding behavior from pre- to post-Flint (Figure 1, panel B). The top row of Figure 2 shows the estimated density of the ninetieth-percentile lead values to the

¹⁷ A PWS is small if it serves 1–3,300 people, medium if it serves 3,301–50,000 people, and large if it serves more than 50,000 people (USEPA 2008).

¹⁸ For additional background, see Supplemental Appendix B.1.

¹⁹ Supplemental Appendix Figures B.1–B.3 plot annual histograms and show a discrete change in reporting behavior in 2016.

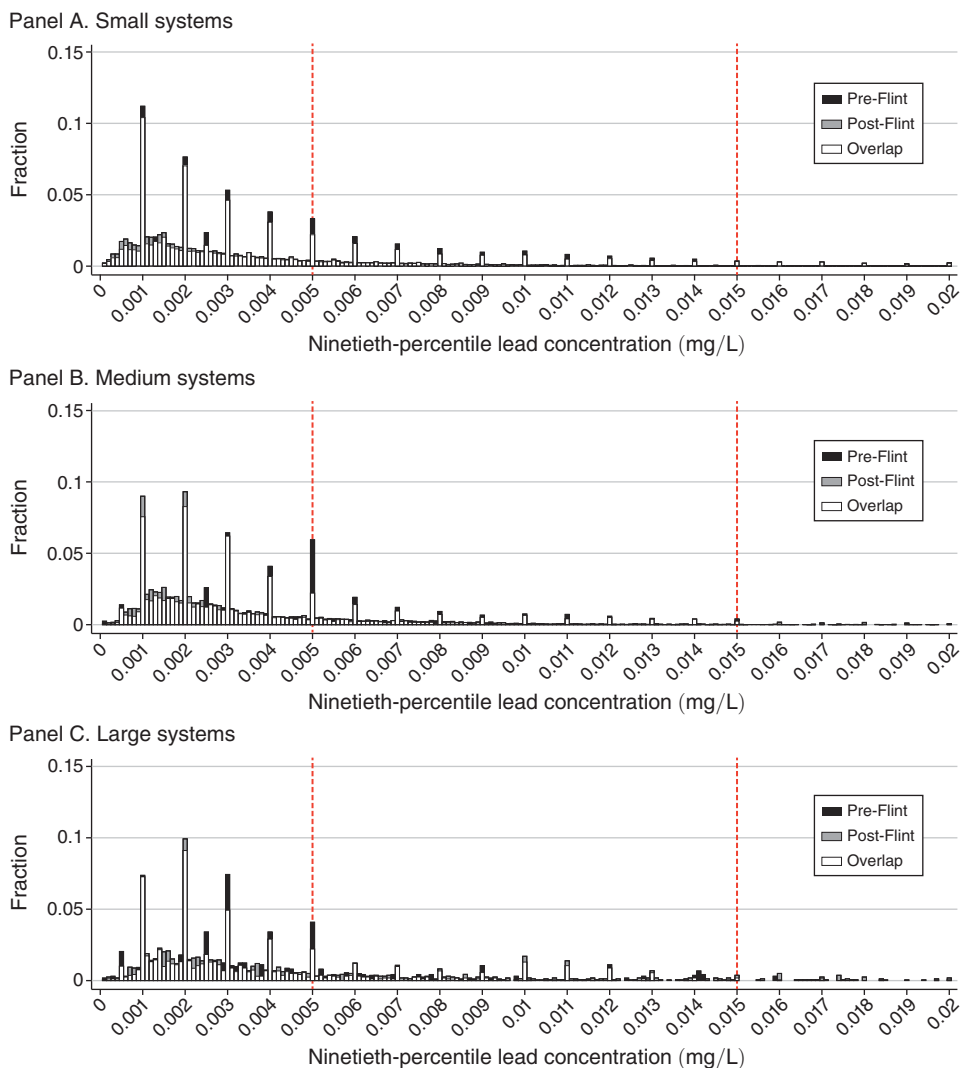


FIGURE 1. HISTOGRAMS OF NINETIETH-PERCENTILE LEAD VALUES BY SYSTEM SIZE, PRE- AND POST-FLINT

Notes: The figures display histograms of reported ninetieth-percentile lead values for small, medium, and large PWSs. Pre-Flint histograms are depicted in black, post-Flint histograms are depicted in gray, and white indicates the overlap of the pre- and post-Flint histograms. The dashed, red vertical lines indicate the reporting threshold (0.005) and action level (0.015).

left and right of the reporting threshold pre- and post-Flint, a visualization of the test statistic developed by McCrary (2008). The McCrary test suggests that threshold manipulation was present and virtually unchanged pre- and post-Flint.²⁰

The bottom row of Figure 2 summarizes the results using our method. The figures show the empirical CDFs of the ninetieth-percentile lead values to the left and

²⁰Supplemental Appendix Table B.1 and Figures B.4–B.8 report results for all system sizes and thresholds.

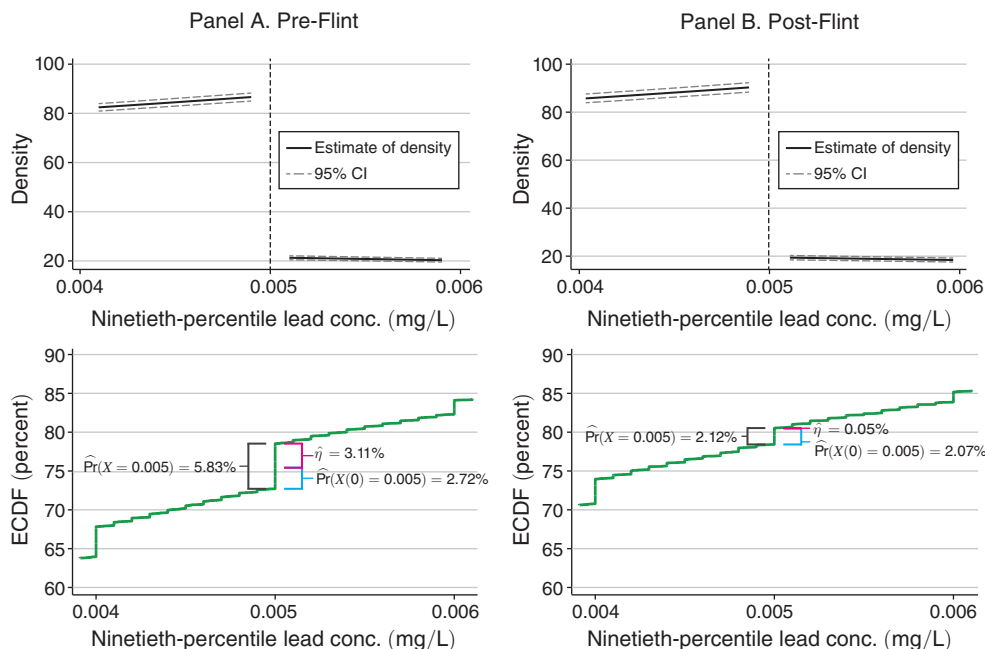


FIGURE 2. MCCRARY TEST AND ILLUSTRATION OF EXCESS ROUNDING—MEDIUM PUBLIC WATER SYSTEMS

Notes: The top row of figures displays the estimated density of ninetyeth-percentile lead concentrations to the left and the right of the reporting threshold (0.005) for medium-sized PWSs pre-Flint (panel A) and post-Flint (panel B). The dashed lines depict the 95 percent confidence interval around the estimate of the density. The bottom row of figures depicts the pre-Flint (panel A) and post-Flint (panel B) empirical CDFs (“ECDF”) (green line), the observed proportions of rounding at the reporting threshold ($\widehat{\Pr}(X = 0.005)$), the counterfactual proportions of rounding (absent manipulation) at the reporting threshold ($\widehat{\Pr}(X(0) = 0.005)$), and estimates of the simple average excess rounding measure at the reporting threshold ($\hat{\eta}^s$).

right of the reporting threshold along with the observed proportion of rounding, $\widehat{\Pr}(X = 0.005)$; the counterfactual (absent manipulation) proportion of rounding, $\widehat{\Pr}(X(0) = 0.005)$; and our proposed manipulation measure, excess rounding ($\hat{\eta}^s$), pre- and post-Flint. Our methods lead to different conclusions. In particular, our excess rounding measure finds evidence consistent with threshold manipulation pre-Flint but not post-Flint.

Table 2 presents our primary results for the reporting threshold (panel A) and the action level (panel B).²¹ Each panel’s first and second rows compare the observed and counterfactual probabilities that a value lies in the roundable interval around each threshold, respectively. The third row presents the difference between these two values, $\hat{\Delta}_j$ from equation (16). The fourth row shows estimates of our simple average excess rounding measure, $\hat{\eta}_j^s$.²²

Starting with the reporting threshold (panel A), we find statistically significant differences between the observed and counterfactual probabilities (i.e., we reject

²¹ Supplemental Appendix Figures B.9–B.11 depict the empirical and counterfactual CDFs obtained via CMLE.

²² We display the probability-weighted average excess rounding ($\hat{\eta}_j^p$) estimates in Supplemental Appendix Figure B.12. The results are similar.

TABLE 2—EXCESS ROUNDING MEASURES AT THE POLICY THRESHOLDS

	Small PWS		Medium PWS		Large PWS	
	Pre-Flint	Post-Flint	Pre-Flint	Post-Flint	Pre-Flint	Post-Flint
<i>Panel A. Reporting threshold ($c_1 = 0.005$)</i>						
(1) $\Pr(l_{c_1} \leq X < u_{c_1})$	0.068	0.058	0.103	0.061	0.084	0.063
(2) $\Pr(l_{c_1} \leq X(0) < u_{c_1})$	0.063	0.056	0.077	0.062	0.079	0.074
Difference: (1)–(2)	0.005	0.002	0.026	–0.001	0.004	–0.011
	0.001	0.001	0.003	0.002	0.008	0.007
η^s (%), homogeneous rounding	0.535 (0.083)	0.070 (0.074)	3.114 (0.261)	0.052 (0.169)	1.260 (0.718)	–0.151 (0.496)
<i>Panel B. Action level ($c_2 = 0.015$)</i>						
(1) $\Pr(l_{c_2} \leq X < u_{c_2})$	0.006	0.006	0.006	0.004	0.005	0.009
(2) $\Pr(l_{c_2} \leq X(0) < u_{c_2})$	0.006	0.006	0.004	0.004	0.005	0.006
Difference: (1)–(2)	0.000	0.000	0.002	0.000	0.000	0.003
	0.000	0.000	0.001	0.001	0.002	0.002
η^s (%), homogeneous rounding	–0.011 (0.028)	0.045 (0.029)	0.162 (0.061)	0.046 (0.056)	–0.052 (0.132)	0.081 (0.155)
Observations	57,774	44,858	13,223	9,908	1,613	1,581

Notes: The table depicts results from our proposed manipulation test and estimates of excess rounding for the reporting threshold of 0.005 mg/L (panel A) and the action level of 0.015 mg/L (panel B) for small, medium, and large PWSs. In each panel, the first row displays estimates of the observed probability of being within the roundable interval ($\widehat{\Pr}(l_j \leq X < u_j)$), the second row displays estimates of the counterfactual probability of being within the roundable interval ($G(u_j; \hat{\theta}) - G(l_j; \hat{\theta})$), and the third row displays estimates of their difference ($\hat{\Delta}_j$ from equation (16)), the numerator of our manipulation test statistic, with $\text{SE}(\hat{\Delta}_j)$ in parentheses. The fourth row of each panel displays estimates of the simple average excess rounding measure ($\hat{\eta}_j^s$). We use manipulation windows of (0.0035, 0.0075) and (0.0135, 0.0175) around the reporting threshold and action level, respectively, to estimate the counterfactual distribution via CMLE. Supplemental Appendix Figures B.9–B.11 present the empirical CDFs and the CMLE counterfactual distribution of $X^N(0)$ used to compute our test and excess rounding measure. We report bootstrapped standard errors in parentheses. For additional details on the specification choices in our implementation, see Supplemental Appendix B.2.

the null hypothesis in equation (4)) of being within the roundable interval around $c_1 = 0.005$ for small and medium systems pre- but not post-Flint. The observed probability of rounding is also higher than the counterfactual for large systems pre-Flint, but the difference is not statistically significant. We find no evidence that the probabilities around the action level (panel B) differ in either period for small or large PWSs. However, medium PWSs show a small difference between the two probabilities pre-Flint but not post-Flint.

The bottom row of each panel in Table 2 reports our excess rounding estimates under the homogeneous rounding assumption. Pre-Flint, a statistically significant 3.11 percent of medium and 0.54 percent of small PWSs' reports exhibited excess rounding at the reporting threshold. Post-Flint, the magnitudes decrease and are not statistically different from zero. For large PWSs, excess rounding is not significantly different from zero in either period. At the action level, excess rounding is much smaller across all system sizes and periods, which is expected since there is much less data around this threshold. Moreover, it is statistically different from zero only for medium PWSs pre-Flint. Figure 3 plots the excess rounding estimates under the homogeneous rounding assumption, their confidence intervals, and the estimated bounds from equation (18). The bounds are relatively tight for small and medium

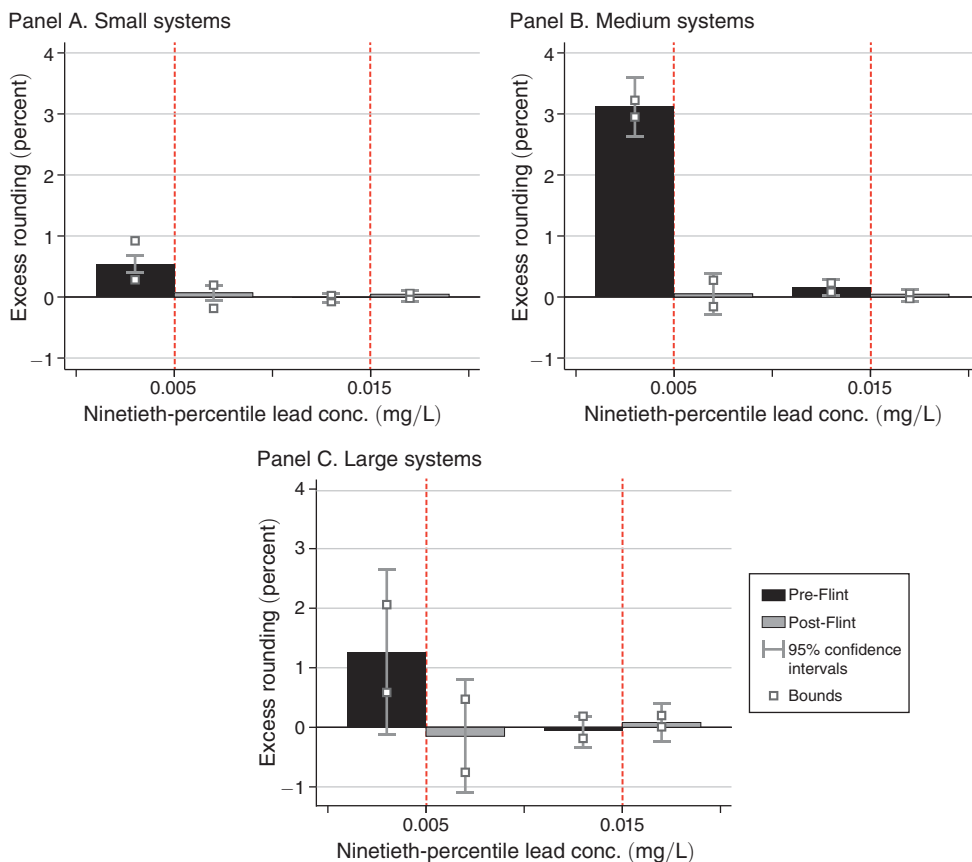


FIGURE 3. EXCESS ROUNDING BY SYSTEM SIZE

Notes: The figures display the simple average excess rounding estimates ($\hat{\eta}_j^s$) for small, medium, and large PWSs. We use manipulation windows of (0.0035, 0.0075) and (0.0135, 0.0175) around the reporting threshold and action level, respectively, to estimate the counterfactual distribution via CMLE. Figures B.9–B.11 present the empirical CDFs and the CMLE counterfactual distribution of $X^N(0)$ used to compute our test and η^s . For additional details on the specification choices in our implementation, see Supplemental Appendix B.2. Pre-Flint estimates are depicted in black; post-Flint estimates are depicted in gray; the estimated bounds from equation (18) are indicated by white, square markers; and the 95 percent confidence intervals for $\hat{\eta}_j^s$ are shown using gray, capped lines.

system sizes, highlighting that rounding away from the policy thresholds was relatively homogeneous.²³

Our results suggest that threshold manipulation was most prominent for small and medium PWSs pre-Flint. Taking the statistically significant excess rounding estimates, we calculate that around 720 of around 71,000 reports by small and medium PWSs were manipulated at the reporting threshold pre-Flint.²⁴ While we cannot detect which PWSs manipulated their reports, we explore the spatial distribution of rounding at the reporting threshold in Supplemental Appendix B.4. The

²³ Supplemental Appendix Table B.2 provides more support for the homogeneous rounding assumption in our setting.

²⁴ We show that our results are not sensitive to a narrower manipulation window (Supplemental Appendix Figure B.13) and conduct several placebo tests by calculating excess rounding at nonpolicy thresholds (Supplemental Appendix Figure B.14).

results suggest that PWSs in one state, Alabama, were responsible for most reports equaling the reporting threshold pre-Flint. Discussions with state regulators suggest that less sensitive lead testing equipment and misapplication of reporting rules were the primary drivers of this behavior.

III. Conclusion

This paper proposes an approach to detect and quantify threshold manipulation in the presence of rounding. Our work suggests several directions for future research. One potentially valuable extension is to use our framework to debias existing discontinuity tests (McCrary 2008; Cattaneo, Jansson, and Ma 2020; Bertanha and Chung 2023). Additionally, the problem of threshold manipulation is closely related to excess bunching, though the primary object of interest in the latter is the elasticity that governs agents' response to a kink or notch in the incentive schedule (e.g., Saez 2010; Chetty et al. 2011; Ito and Sallee 2018). Our assumptions on threshold manipulation include the specification of manipulation windows and a flexible parametric family of distributions to achieve point identification. A fruitful direction for future work could build on recent work in the excess bunching literature and provide more robust bounding approaches that relax these restrictions (Blomquist et al. 2021; Bertanha, McCallum, and Seegert 2023).

Our work further demonstrates that allowing rounding to policy thresholds limits the types of threshold manipulation one can quantify. We only quantify threshold manipulation that leads to mass at the policy threshold. In our application, this could have been avoided if the EPA did not allow PWSs to round their reported data to the same number of digits as the policy thresholds. A natural policy implication of our findings is that regulators should only allow rounding to a decimal point greater than the policy threshold, though the EPA may need to require PWSs to use laboratories with more precise equipment than many have used historically for such a requirement to be feasible. Finally, one interesting direction for future work could explore how these practices impacted public health and lead exposure.

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